

LeanForControl

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Chapter 1

Linear systems

1.1 Observability and controllability matrices

For a finite-dimensional linear system

$$\dot{x} = Ax + Bu, \quad y = Cx,$$

with state matrix A , input matrix B , and output matrix C , this section collects the (finite-horizon) controllability and observability matrices and the bridge theorem connecting the textbook observability predicate to the kernel of the observability matrix.

1.1.1 Observability matrix

Definition 1. The *observability matrix* of a pair (A, C) with $A \in \mathbb{F}^{n \times n}$ and $C \in \mathbb{F}^{p \times n}$ is the block-row matrix

$$\mathcal{O}(A, C) = \begin{bmatrix} C \\ C A \\ C A^2 \\ \vdots \\ C A^{n-1} \end{bmatrix} \in \mathbb{F}^{(np) \times n}.$$

Rows are indexed by $\text{Fin } n \times \text{Fin } p$, so that A^k is available without casting $k : \text{Fin } n$ through Fin.val .

Definition 2. A linear system (A, C) is *observable* when the only state $x \in \mathbb{F}^n$ for which

$$C A^k x = 0 \quad \text{for every } k = 0, 1, \dots, n-1$$

is the zero state. This phrasing does not mention $\mathcal{O}(A, C)$, so the bridge theorem 8 has real content.

Theorem 3. *Block-row entry shape: at row (k, i) and column j , the observability matrix coincides with the (i, j) entry of $C A^k$:*

$$\mathcal{O}(A, C)_{(k,i),j} = (C A^k)_{i,j}.$$

Proof. Holds definitionally by the encoding chosen for definition 1. □

Theorem 4. For every state $x \in \mathbb{F}^n$ and every $(k, i) \in \text{Fin } n \times \text{Fin } p$,

$$(\mathcal{O}(A, C) \cdot x)_{(k, i)} = (C A^k \cdot x)_i.$$

This is the definitional bridge that powers theorem 8.

Proof. Holds by definitional unfolding of $\mathcal{O}(A, C)$ (see definition 1). $\square$

1.1.2 Controllability matrix

Definition 5. The *controllability matrix* of a pair (A, B) with $A \in \mathbb{F}^{n \times n}$ and $B \in \mathbb{F}^{n \times m}$ is the block-column matrix

$$\mathcal{C}(A, B) = [B \quad AB \quad A^2 B \quad \dots \quad A^{n-1} B] \in \mathbb{F}^{n \times (nm)}.$$

Columns are indexed by $\text{Fin } n \times \text{Fin } m$, so that A^k is available without casting $k : \text{Fin } n$ through Fin.val .

Theorem 6. Block-column entry shape: at row i and column (k, j) , the controllability matrix coincides with the (i, j) entry of $A^k B$:

$$\mathcal{C}(A, B)_{i, (k, j)} = (A^k B)_{i, j}.$$

Proof. Holds definitionally by the encoding chosen for definition 5. $\square$

Definition 7. A linear system (A, B) is *controllable* when every target state $x \in \mathbb{F}^n$ is reachable from the origin in n steps: there exist input vectors $u_0, u_1, \dots, u_{n-1} \in \mathbb{F}^m$ such that

$$x = \sum_{k=0}^{n-1} A^k B u_k.$$

This phrasing does not name the controllability matrix, so the bridge theorem 10 has real content.

1.1.3 Milestone: observability iff trivial kernel of $\mathcal{O}$

Theorem 8. A finite-dimensional linear system (A, C) is observable in the sense of definition 2 if and only if the observability matrix $\mathcal{O}(A, C)$ has trivial kernel under matrix-vector multiplication:

$$\text{IsObservable}(A, C) \iff (\forall x, \mathcal{O}(A, C) \cdot x = 0 \Rightarrow x = 0).$$

Proof. Both directions follow from the bridge theorem 4, which identifies the (k, i) -coordinate of $\mathcal{O}(A, C) \cdot x$ with the i -coordinate of $(C A^k) \cdot x$. The forward direction reads the per-power kernel out of the assembled kernel coordinatewise; the reverse direction packs them back. $\square$

1.1.4 Rank characterizations

Theorem 9. A finite-dimensional system (A, C) is observable if and only if the observability matrix $\mathcal{O}(A, C)$ has full column rank, i.e.

$$\text{rank } \mathcal{O}(A, C) = n,$$

where n is the state dimension.

Proof. Chain the kernel-form milestone theorem 8 with the matrix-level bridge $(\forall x, M \cdot x = 0 \Rightarrow x = 0) \iff \text{rank } M = n$ from ‘MatrixAlgebra.Rank’, applied with $M = \mathcal{O}(A, C)$. $\square$

Theorem 10. *A linear system (A, B) is controllable if and only if the controllability matrix has full row rank,*

$$\text{rank } \mathcal{C}(A, B) = n,$$

where n is the state dimension.

Proof. Combine the reindexing identity $\mathcal{C}(A, B) \cdot u = \sum_k A^k B \cdot u_k$ (with $u_k(j) = u(k, j)$) with the matrix-level bridge $(\forall y, \exists x, Mx = y) \iff \text{rank } M = n$ from ‘MatrixAlgebra.Rank’, applied with $M = \mathcal{C}(A, B)$. $\square$

1.1.5 Hautus observability

Definition 11. The *unobservable subspace* of (A, C) is the A -invariant subspace

$$\mathcal{N}(A, C) = \{v \in \mathbb{C}^n : C A^k v = 0 \text{ for every } k = 0, 1, \dots, n-1\}.$$

Equivalently, $\mathcal{N}(A, C) = \ker \mathcal{O}(A, C)$, but here it is phrased without naming $\mathcal{O}$ so the bridge theorem 12 reads as content.

Theorem 12. *A finite-dimensional system (A, C) is observable in the sense of definition 2 if and only if its unobservable subspace is trivial:*

$$\mathcal{N}(A, C) = \{0\} \iff \text{IsObservable}(A, C).$$

Proof. Both directions are membership unfoldings of definition 11 against definition 2. $\square$

Theorem 13. *The unobservable subspace is closed under the action of A : for every $v \in \mathcal{N}(A, C)$, also $Av \in \mathcal{N}(A, C)$.*

Proof. For $k = 0, \dots, n-2$ this is direct from the definition. For $k = n-1$ we land at $CA^{n-1}v$, which Cayley-Hamilton rewrites as a $\mathbb{C}$ -linear combination of $CA^i v$ for $i < n$. Each of those is zero, so the combination is zero. $\square$

Definition 14. The *Hautus observability matrix* of (A, C) at a complex number μ is the block-row matrix

$$H_{A,C}(\mu) = \begin{bmatrix} \mu I - A \\ C \end{bmatrix} \in \mathbb{C}^{(n+p) \times n}.$$

Theorem 15. *For every $\mu \in \mathbb{C}$ and $v \in \mathbb{C}^n$,*

$$H_{A,C}(\mu) \cdot v = 0 \iff (Av = \mu v \wedge Cv = 0).$$

Proof. $\square$

Theorem 16. *If (A, C) is not observable, then there exists $\mu \in \mathbb{C}$ and a nonzero vector $v \in \mathbb{C}^n$ such that*

$$H_{A,C}(\mu) \cdot v = 0,$$

i.e. μ is an eigenvalue of A with an eigenvector in $\ker C$.

Proof. The unobservable subspace definition 11 is nontrivial by theorem 12. It is A -invariant by theorem 13, hence finite-dimensional and invariant; over $\mathbb{C}$ this guarantees a nonzero eigenvector inside it. That vector is automatically in $\ker C$ (membership at $k = 0$), so it lies in the kernel of $H_{A,C}(\mu)$ via theorem 15. $\square$

Theorem 17. *If $\mu \in \mathbb{C}$ and $v \neq 0$ satisfy $H_{A,C}(\mu) \cdot v = 0$, then (A, C) is not observable.*

Proof. The witness gives $Av = \mu v$ and $Cv = 0$. By induction on k , $A^k v = \mu^k v$, so $C A^k v = \mu^k Cv = 0$ for every k . The vector v is therefore in the unobservable subspace and is nonzero, so observability fails. $\square$

Theorem 18. *A finite-dimensional system (A, C) over $\mathbb{C}$ is observable if and only if for every $\mu \in \mathbb{C}$ the Hautus matrix $H_{A,C}(\mu)$ has trivial kernel:*

$$\text{IsObservable}(A, C) \iff \forall \mu \in \mathbb{C}, \ker H_{A,C}(\mu) = \{0\}.$$

Proof. Combine theorem 16 (failure direction) and theorem 17 (converse). $\square$

1.1.6 Hautus controllability

Definition 19. The *Hautus controllability matrix* of (A, B) at a complex number μ is the block-column matrix

$$H_{A,B}^{\text{ctrl}}(\mu) = [\mu I - A \quad B] \in \mathbb{C}^{n \times (n+m)}.$$

Theorem 20. *A finite-dimensional system (A, B) over $\mathbb{C}$ is controllable if and only if for every $\mu \in \mathbb{C}$ the Hautus matrix $H_{A,B}^{\text{ctrl}}(\mu)$ has full row rank:*

$$\text{IsControllable}(A, B) \iff \forall \mu \in \mathbb{C}, \text{rank } H_{A,B}^{\text{ctrl}}(\mu) = n.$$

Proof. Combine the duality bridge `isControllable_iff_isObservable_transpose` with the observability Hautus iff theorem 18, then convert the kernel-form RHS to the rank form via the matrix bridge `mulVec_kernel_trivial_iff_rank_eq_card_cols` from ‘MatrixAlgebra.Rank’, and finally identify the transposed Hautus matrices. $\square$

1.2 Reachable subspace and Kalman decomposition

Definition 21. The reachable subspace of a pair (A, B) is the column span of its controllability matrix:

$$\mathcal{R}(A, B) = \text{range } \mathcal{C}(A, B).$$

Thus it is exactly the span of $B, AB, \dots, A^{n-1}B$.

Theorem 22. *Let $A \in \mathbb{F}^{n \times n}$ and $B \in \mathbb{F}^{n \times m}$, and let $\mathcal{C}(A, B) \in \mathbb{F}^{n \times (nm)}$ be their controllability matrix. For $x \in \mathbb{F}^n$,*

$$x \in \mathcal{R}(A, B) \iff \exists u : \text{Fin}(n) \times \text{Fin}(m) \rightarrow \mathbb{F}, \quad \mathcal{C}(A, B)u = x.$$

Here $\mathcal{R}(A, B) = \text{range } \mathcal{C}(A, B)$ is the reachable subspace.

Proof. $\square$

Theorem 23. *A pair (A, B) is controllable if and only if its reachable subspace is the whole state space.*

Proof. □

Theorem 24. *The reachable subspace is A -invariant: $A\mathcal{R}(A, B) \subseteq \mathcal{R}(A, B)$.*

Proof. □

Definition 25. A Kalman decomposition of (A, B, C) records four coordinate sectors adapted to the reachable and unobservable subspaces. Their direct sums recover those subspaces and the full state space. The chosen complements are noncanonical and are not individually asserted to be invariant under A .

Theorem 26. *Every finite-dimensional complex state-space system has four coordinate sectors adapted to its reachable and unobservable subspaces. The complements are noncanonical vector-space complements and are not individually asserted to be invariant under A .*

Proof. □

Definition 27. Addition of the four Kalman coordinate sectors defines a linear equivalence $X_{c\bar{o}} \times X_{co} \times X_{\bar{c}\bar{o}} \times X_{\bar{c}o} \simeq \mathbb{C}^n$. Choosing sector bases therefore gives an adapted basis of the original state space.

Theorem 28. *In a basis adapted to the four Kalman coordinate sectors, ordered as controllable-unobservable, controllable-observable, uncontrollable-unobservable, uncontrollable-observable, the system has the forced-zero pattern*

$$A' = \begin{bmatrix} * & * & * & * \\ 0 & * & 0 & * \\ 0 & 0 & * & * \\ 0 & 0 & 0 & * \end{bmatrix}, \quad B' = \begin{bmatrix} * \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad C' = [0 \quad * \quad 0 \quad *].$$

Every displayed zero is asserted entrywise; no vanishing claim is made for the starred blocks. The chosen complement sectors are not individually asserted to be invariant under A .

Proof. Reachable-subspace invariance forces the lower-left state and input zeros, unobservable-subspace invariance forces the remaining state zeros, and the zeroth observability condition forces the output zeros. Coordinate-basis representation lemmas translate these component statements to individual matrix entries. □

1.3 Hurwitz stability

For a real square matrix A , Hurwitz stability is expressed through the complex eigenpairs of its entrywise complexification. The rate-indexed predicate records a strict spectral margin, and ordinary Hurwitz stability is the zero-rate case.

Definition 29. A real square matrix A is *Hurwitz with decay rate α* when every complex eigenpair (μ, v) with $v \neq 0$ satisfies $\text{Re}(\mu) < -\alpha$.

Definition 30. A real square matrix is *Hurwitz* when every complex eigenvalue has strictly negative real part.

Theorem 31. A real matrix A is Hurwitz with decay rate α if and only if the spectrally shifted matrix $A + \alpha I$ is Hurwitz.

Proof. A complex eigenvalue μ of A becomes $\mu + \alpha$ after the shift, and $\operatorname{Re}(\mu) < -\alpha$ is equivalent to $\operatorname{Re}(\mu + \alpha) < 0$. $\square$

The exponential decay estimates rest on a general Banach-algebra fact, stated with no matrices in it: an element of spectral radius below one has a contractive power.

1.4 Matrix Lyapunov equations and quadratic certificates

1.5 Peano–Baker series and the state transition matrix

For a genuinely time-varying state matrix $A(t)$, this section constructs the state transition matrix $\Phi(t, t_0)$ as the Peano–Baker series and proves Theorem 5.1: Φ solves the matrix ODE $\dot{\Phi}(t, t_0) = A(t)\Phi(t, t_0)$ with $\Phi(t_0, t_0) = I$, and consequently $x(t) := \Phi(t, t_0)x_0$ solves $\dot{x} = A(t)x$, $x(t_0) = x_0$.

Standing hypotheses. Throughout this section A is continuous and satisfies $\|A(s)\| \leq M$ for all s on the segment between the two times named in each statement. Both are needed: continuity to differentiate the series term by term, the bound to make it converge at all. Statements below do not repeat them.

1.5.1 Uniqueness

1.5.2 Semigroup, invertibility and composition

Uniqueness is what makes these available: two expressions that solve the same matrix initial value problem must agree, and each identity below is an instance of that.

1.5.3 Variation of constants

With a forcing term $B(t)u(t)$, the solution acquires a convolution against Φ . The difficulty is that the elapsed time appears both as the upper limit of integration and inside the integrand; the proof factors the two apart rather than differentiating Φ in its second argument.

1.6 Discrete-time systems

For $x(t+1) = A(t)x(t)$ the state transition matrix is an ordered product rather than a series, and the whole theory becomes finite: every proof below is an induction, with no convergence, no differentiation and no Gronwall estimate. Two differences from the continuous case are substantive rather than merely technical. There is no standing bound on A to carry, since a finite product always converges; and Φ need not be invertible, so the semigroup property genuinely requires $\tau \leq s \leq t$ instead of holding in any order.

1.6.1 Discrete variation of constants

1.6.2 The LTI case: the matrix exponential

For a constant state matrix the Peano–Baker series sums to the matrix exponential, which is Hespanha’s (6.1)–(6.2). The point of proving it is economy: every property of Φ established above for a time-varying A then holds for $e^{A(t-t_0)}$ by specialization, with no separate LTI theory. The two exponential facts that need no solution theory at all — the semigroup law (P6.3) and the finite-polynomial form via Cayley–Hamilton (P6.5) — are proved as statements about $\exp$ and A alone.

1.6.3 The matrix exponential as an algebraic object

These say nothing about trajectories, initial times or state transition matrices — they are facts about $\exp$ and A alone, and are proved as such. The complexification bridge is what lets spectral arguments, which must happen over $\mathbb{C}$, yield bounds on a real exponential; P6.5 collapses the exponential series to a finite polynomial in A .

Chapter 2

Ordinary differential equations

2.1 Continuity toolbox

Two facts about continuous real functions and curves, with no differential-equation content of their own, used throughout the chapters that follow: the comparison lemma needs the last root before a sign change, and the Lyapunov stability arguments need the first time a curve reaches a sphere.

2.2 Interval integrals

Estimates and integrability side conditions for the interval integral, used wherever a differential equation is handled in integral form. The last three concern an integral whose endpoints move: normalizing by the window length makes it a mean value, after which continuity, antitonicity and decay each need only a hypothesis on the endpoints themselves. They support the class $\mathcal{KL}$ construction in the last chapter.

2.3 Limits superior

Real-valued $\limsup$ is an infimum of upper bounds, so it is only well-behaved where the function is bounded. The facts collected here are stated over an arbitrary filter and have no differential-equation content; the Dini derivative of the next section is the only current consumer.

2.4 Dini derivatives

The upper right Dini derivative replaces the ordinary derivative where a function is only known to be continuous, which is the situation the comparison lemma is in. It is defined here as a $\limsup$ valued in $\mathbb{R}$ rather than in $[-\infty, +\infty]$, so most results below carry a hypothesis that the forward difference quotients are bounded: without it an unbounded quotient makes the $\limsup$ an infimum over the empty set, which $\mathbb{R}$ reports as 0.

Definition 32. The *upper right Dini derivative* of $f : \mathbb{R} \rightarrow \mathbb{R}$ at t is

$$D^+f(t) = \limsup_{h \rightarrow 0^+} \frac{f(t+h) - f(t)}{h}.$$

When f is differentiable at t this equals $f'(t)$.

2.4.1 Relationship to the classical derivative

2.4.2 Characterization by difference quotients

2.4.3 Linearity properties

2.4.4 The comparison step

2.5 Gronwall–Bellman inequality

A function obeying an integral inequality is bounded by the solution of the corresponding integral equation. The two specializations are what the rest of the development actually calls: constant Λ gives exponential growth, and constant Λ and μ gives the $Ce^{\mu(t-a)}$ bound behind every continuous-dependence estimate.

Theorem 33. Gronwall–Bellman inequality. Let $\Lambda, \mu : [a, b] \rightarrow \mathbb{R}$ be continuous with $\mu \geq 0$, and let $y : [a, b] \rightarrow \mathbb{R}$ be continuous satisfying

$$y(t) \leq \Lambda(t) + \int_a^t \mu(s) y(s) \, ds \quad \forall t \in [a, b].$$

Then

$$y(t) \leq \Lambda(t) + \int_a^t \Lambda(s) \mu(s) e^{\int_s^t \mu(\tau) \, d\tau} \, ds \quad \forall t \in [a, b].$$

Proof.

□

2.6 Continuous dependence on parameters

Definition 34. A function $x : [t_0, t_1] \rightarrow E$ is an *integral solution* of $\dot{x} = F(t, x)$ with initial value x_0 when

$$x(t) = x_0 + \int_{t_0}^t F(s, x(s)) \, ds \quad \forall t \in [t_0, t_1].$$

2.6.1 The integral and differential formulations

An ODE can be written as a differential equation, $\dot{x} = F(t, x)$, or as the equivalent integral equation. The continuous-dependence and Gronwall arguments want the integral form, which presupposes no differentiability and carries its initial condition; the Lyapunov arguments want the differential form, so that $V \circ \varphi$ can be differentiated. The results below are the two directions of the fundamental theorem of calculus relating them.

They are provision rather than current practice: nothing in the stability chapters has yet been routed through this equivalence, so it is the intended path from a trajectory to the results of this chapter rather than an exercised one.

2.6.2 Time invariance and local existence

An autonomous field has no preferred time origin, which is what lets the autonomous stability predicates fix the initial time at 0.

2.6.3 Continuous dependence

Two solutions of the same equation, started at different points or subject to different perturbations, separate no faster than exponentially — at a rate fixed by the Lipschitz constant of the field. Everything downstream that needs uniqueness gets it from here, by taking the two solutions to agree initially and the perturbation to vanish, which collapses the bound to zero.

Theorem 35. *Theorem 3.5 (Continuous dependence on parameters). If y solves $\dot{y} = f(t, y)$ and z solves $\dot{z} = f(t, z) + g(t, z)$ with $\|g(t, x)\| \leq \alpha$ and $\|z_0 - y_0\| \leq \alpha$, and $\alpha(1 + 1/L)e^{L(t_1 - t_0)} \leq \varepsilon$, then $\|y(t) - z(t)\| \leq \varepsilon$ for all $t \in [t_0, t_1]$.*

Proof. Reduce to theorem 33 applied to $\|y - z\|$, using the L -Lipschitz bound on f and the α -bound on g . $\square$

2.7 Comparison lemma

The comparison lemma bounds a function satisfying a differential *inequality* by the solution of the corresponding differential *equation*. The inequality is stated with the Dini derivative, so v need only be continuous — which is what makes the lemma applicable to $\|x(t)\|$, not differentiable where a trajectory crosses the origin.

Theorem 36. *Comparison Lemma 3.4. If u solves $\dot{u} = f(t, u)$ with $u(t_0) = u_0$, and v is continuous with $D^+v(t) \leq f(t, v(t))$ and $v(t_0) \leq u_0$, then $v(t) \leq u(t)$ for all $t \in [t_0, t_1]$.*

Proof. For each $\lambda > 0$, the perturbed solution z_λ of $\dot{z} = f(t, z) + \lambda$ satisfies $v \leq z_\lambda$ by ???. By theorem 35, $\|u - z_\lambda\| \leq \varepsilon/2$. Since $\varepsilon > 0$ is arbitrary, $v(t) \leq u(t)$. $\square$

Chapter 3

Lyapunov stability

3.1 Trajectories, equilibria, and comparison functions

Definition 37. A point $x_{\text{eq}} \in \mathbb{R}^n$ is an *equilibrium* of $\dot{x} = f(x)$ when $f(x_{\text{eq}}) = 0$.

Definition 38. A *class $\mathcal{K}$* function on $[0, a)$ is a continuous strictly increasing map $\alpha : [0, a) \rightarrow [0, b)$ with $\alpha(0) = 0$. The structure records both α and its inverse $\alpha^{-1} : [0, b) \rightarrow [0, a)$.

Definition 39. A *class $\mathcal{K}_\infty$* function is a continuous strictly increasing map $\alpha : [0, \infty) \rightarrow [0, \infty)$ with $\alpha(0) = 0$ and $\alpha(r) \rightarrow \infty$ as $r \rightarrow \infty$.

3.1.1 Building class $\mathcal{K}$ functions

A recognition criterion and the closure properties. The restatements of the definition itself (strict monotonicity as an iff, the two inverse equations, and so on) are left to the Lean API; they say nothing the definition above does not.

3.1.2 Building class $\mathcal{K}_\infty$ functions

The same recognition criterion and closure properties on the unbounded domain, together with the closure properties that need the global domain (sum, positive scaling, minimum), the power functions as the standard examples, and the restriction back to class $\mathcal{K}$.

3.2 Stability predicates

The predicates below quantify over every solution segment on a compact interval $[t_0, t_1]$. This prevents finite escape from making a stability claim vacuous merely because a non-equilibrium solution is not globally defined. A solution segment is an integral curve of the vector field restricted to that interval, in the sense of `IsTrajectoryOn`.

Definition 40. The equilibrium x_{eq} is *Lyapunov stable* when

$$\forall \varepsilon > 0, \exists \delta > 0, \forall \varphi, \text{IsTrajectory}(\varphi, f) \Rightarrow \|\varphi(0) - x_{\text{eq}}\| < \delta \Rightarrow \forall t \geq 0, \|\varphi(t) - x_{\text{eq}}\| < \varepsilon.$$

Definition 41. The equilibrium x_{eq} is *locally asymptotically stable* (LAS) when it is Lyapunov stable (definition 40) and there exists $c > 0$ such that every trajectory φ with $\|\varphi(0) - x_{\text{eq}}\| < c$ satisfies $\varphi(t) \rightarrow x_{\text{eq}}$ as $t \rightarrow \infty$.

Definition 42. The equilibrium x_{eq} is *globally asymptotically stable* (GAS) when it is Lyapunov stable (definition 40) and every trajectory φ satisfies $\varphi(t) \rightarrow x_{\text{eq}}$ as $t \rightarrow \infty$.

3.3 Sublevel sets and positive invariance

Definition 43. The *sublevel set* of $V : \mathbb{R}^n \rightarrow \mathbb{R}$ at level $c \in \mathbb{R}$ is

$$\Omega_c(V) = \{x \in \mathbb{R}^n : V(x) \leq c\}.$$

Definition 44. A set $S \subseteq \mathbb{R}^n$ is *positively invariant* for $\dot{x} = f(x)$ when every trajectory φ starting in S remains in S for all future time:

$$\varphi(0) \in S \Rightarrow \varphi(t) \in S \quad \forall t \geq 0.$$

Theorem 45. If $V : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuous and radially unbounded ($V(x) \rightarrow \infty$ as $\|x\| \rightarrow \infty$), then every sublevel set $\Omega_c(V)$ (definition 43) is compact.

Proof. Closed: $\Omega_c(V) = V^{-1}((-\infty, c])$ by continuity. Bounded: coercivity yields R with $\Omega_c(V) \subseteq \overline{B}(0, R)$. Compact: Heine–Borel in $\mathbb{R}^n$. $\square$

3.4 Lyapunov functions

Definition 46. A function $V : \mathbb{R}^n \rightarrow \mathbb{R}$ is a *local Lyapunov function* on an open domain $D \ni x_{\text{eq}}$ when V is smooth, $V(x_{\text{eq}}) = 0$, $V > 0$ on $D \setminus \{x_{\text{eq}}\}$, and the Lie derivative satisfies $\dot{V}(x) = \nabla V(x) \cdot f(x) \leq 0$ for all $x \in D$.

Definition 47. A *strict local Lyapunov function* on D strengthens definition 46: the Lie derivative satisfies $\dot{V}(x) < 0$ for all $x \in D \setminus \{x_{\text{eq}}\}$, and there exists $c > 0$ such that the compact sublevel set $\Omega_c(V) \subseteq D$ (definition 43).

Definition 48. A *global strict Lyapunov function* for $\dot{x} = f(x)$ at x_{eq} is a C^1 map $V : \mathbb{R}^n \rightarrow \mathbb{R}$ with $V(x_{\text{eq}}) = 0$, $V > 0$ everywhere else, $\dot{V}(x) < 0$ on $\mathbb{R}^n \setminus \{x_{\text{eq}}\}$, and all sublevel sets $\Omega_c(V)$ compact (coercivity).

Definition 49. The classical GAS Lyapunov certificate: a C^1 map $V : \mathbb{R}^n \rightarrow \mathbb{R}$ with $V(x_{\text{eq}}) = 0$, $V > 0$ elsewhere, $\dot{V} < 0$ on $\mathbb{R}^n \setminus \{x_{\text{eq}}\}$, and radially unbounded ($V(x) \rightarrow \infty$ as $\|x\| \rightarrow \infty$). Implies definition 48 via theorem 45.

3.4.1 The hierarchy of certificates

The four notions above differ only in how much they demand; each stronger one weakens to the next, so a theorem proved for the weakest applies to all of them.

3.5 Differentiating along solutions

Every Lyapunov argument begins the same way: compose the certificate with a solution and differentiate. The chain rule gives the Lie derivative $DV(\varphi(t))[f(\varphi(t))]$, and a sign condition on it becomes a monotonicity statement about $V \circ \varphi$. Collecting that step here fixes one spelling of the Lie derivative for the whole development.

3.6 Milestone theorems for autonomous systems

Theorem 50. *Lyapunov's stability theorem.* *If V is a local Lyapunov function (definition 46) for $\dot{x} = f(x)$ on a domain $D \ni x_{\text{eq}}$, then x_{eq} is Lyapunov stable (definition 40).*

Proof. Pick ε_0 so $\overline{B}(x_{\text{eq}}, \varepsilon_0) \subseteq D$. Let $m = \min_{S_{\varepsilon_0}} V > 0$. Choose δ with $V < m$ on $B(x_{\text{eq}}, \delta)$. If $\|\varphi(t^*) - x_{\text{eq}}\| \geq \varepsilon$, monotonicity of V gives $V(\varphi(t^*)) \leq V(\varphi(0)) < m \leq V(\varphi(t^*))$, a contradiction. $\square$

Theorem 51. *Lyapunov's local asymptotic stability theorem.* *If V is a strict local Lyapunov function (definition 47) and f is continuous, then x_{eq} is locally asymptotically stable (definition 41).*

Proof. The compact sublevel set $\Omega_{c_0} \subseteq D$ is positively invariant. $V(\varphi(t)) \rightarrow L \geq 0$ by monotone convergence; $L = 0$ by compactness; $\varphi(t) \rightarrow x_{\text{eq}}$. $\square$

Theorem 52. *Lyapunov's global asymptotic stability theorem.* *If V is a global strict Lyapunov function (definition 48) and f is continuous, then x_{eq} is globally asymptotically stable (definition 42).*

Proof. Stability from theorem 50. For convergence: $V(\varphi(t)) \rightarrow L \geq 0$ by monotone convergence; $L = 0$ by a LaSalle-type argument; $\varphi(t) \rightarrow x_{\text{eq}}$ by coercivity. $\square$

Theorem 53. *Corollary.* *If V is a radially unbounded strict Lyapunov function (definition 49) and f is continuous, then x_{eq} is globally asymptotically stable.*

Proof. Radial unboundedness gives compact sublevel sets (theorem 45), so V satisfies definition 48; apply theorem 52. $\square$

The asymptotic results rest on a uniformity that mere per-solution convergence does not give: outside a ball about the equilibrium the Lie derivative is bounded away from zero, so every solution is drained at a shared rate.

3.6.1 Stability from exponential decay, and instability from escape

3.7 Lyapunov's indirect method

The stable-branch theorem gives uniform local exponential decay on every finite forward solution segment, without silently excluding solutions that are not globally defined.

3.8 LaSalle's invariance principle

Theorem 54. *LaSalle's invariance principle.* Let Ω be compact and positively invariant for $\dot{x} = f(x)$, $V \in C^1$ with $\dot{V}(x) \leq 0$ on Ω , and $M \subseteq \Omega$ closed. If the ω -limit set of any trajectory φ starting in Ω satisfies $\omega(\varphi) \subseteq M$, then $\varphi(t) \rightarrow M$ as $t \rightarrow \infty$.

Proof. □

Theorem 55. *Barbashin's theorem.* If $V \in C^1$ is positive definite on D , $\dot{V} \leq 0$ on D , there exists a compact sublevel set $\Omega_c \subseteq D$, and every trajectory starting in Ω_c has ω -limit set $\subseteq \{x_{\text{eq}}\}$, then x_{eq} is locally asymptotically stable (definition 41).

Proof. □

Theorem 56. *Krasovskii's theorem.* If $V \in C^1$ is positive definite, $\dot{V} \leq 0$ on $\mathbb{R}^n$, radially unbounded, and every trajectory has ω -limit set $\subseteq \{x_{\text{eq}}\}$, then x_{eq} is globally asymptotically stable (definition 42).

Proof. □

3.9 Lyapunov class $\mathcal{K}$ bounds

Theorem 57. *Class $\mathcal{K}$ sandwich bounds.* For any continuous positive-definite function $V : \mathbb{R}^n \rightarrow \mathbb{R}$ on $\overline{B}(0, r)$, there exist class $\mathcal{K}$ functions α_1, α_2 such that

$$\alpha_1(\|x\|) \leq V(x) \leq \alpha_2(\|x\|) \quad \forall \|x\| \leq r.$$

Proof. Construct α_1 from the infimum of V on annuli (lower bound), and α_2 from the supremum of V on balls (upper bound). Each is sandwiched by a class $\mathcal{K}$ function via the axioms in `LeanForControl.axioms`. □

Chapter 4

Non-autonomous systems

For $\dot{x} = f(t, x)$ the initial time matters, and the definitions split accordingly: a bound may hold at each initial time with a margin depending on it, or with one margin serving every initial time. That distinction — plain versus *uniform* — is what the whole chapter turns on, and it has no counterpart in the autonomous theory, where time invariance lets the initial time be fixed at 0 without loss.

4.1 Trajectories and equilibria

Definition 58. A *trajectory* of the non-autonomous ODE $\dot{x} = f(t, x)$ is a globally defined map $\varphi : \mathbb{R} \rightarrow \mathbb{R}^n$ satisfying

$$\dot{\varphi}(t) = f(t, \varphi(t)) \quad \text{for every } t \in \mathbb{R}.$$

Definition 59. A point $x_{\text{eq}} \in \mathbb{R}^n$ is an *equilibrium* of the non-autonomous ODE $\dot{x} = f(t, x)$ when $f(t, x_{\text{eq}}) = 0$ for every $t \in \mathbb{R}$.

4.2 Uniform convergence times and comparison-function bounds

The estimates the class $\mathcal{KL}$ characterization is assembled from. Naming them here, rather than re-typing their quantifiers at each use, is what makes the characterization statements readable.

4.3 Stability predicates

Definition 60. The equilibrium x_{eq} of $\dot{x} = f(t, x)$ is *stable* when

$$\forall \varepsilon > 0, \forall t_0 \geq 0, \exists \delta = \delta(\varepsilon, t_0) > 0, \forall \varphi, \text{IsTrajectoryNA}(\varphi, f) \Rightarrow \|\varphi(t_0) - x_{\text{eq}}\| < \delta \Rightarrow \forall t \geq t_0, \|\varphi(t) - x_{\text{eq}}\| < \varepsilon.$$

Definition 61. The equilibrium x_{eq} is *uniformly stable* when δ in definition 60 can be chosen independently of t_0 :

$$\forall \varepsilon > 0, \exists \delta = \delta(\varepsilon) > 0, \forall t_0 \geq 0, \forall \varphi, \text{IsTrajectoryNA}(\varphi, f) \Rightarrow \|\varphi(t_0) - x_{\text{eq}}\| < \delta \Rightarrow \forall t \geq t_0, \|\varphi(t) - x_{\text{eq}}\| < \varepsilon.$$

Definition 62. The equilibrium x_{eq} is *unstable* when it is not stable (definition 60).

4.4 Asymptotic and exponential stability

Definition 63. The equilibrium x_{eq} is *asymptotically stable* when it is stable (definition 60) and for each $t_0 \geq 0$ there exists $c = c(t_0) > 0$ such that every trajectory φ with $\|\varphi(t_0) - x_{\text{eq}}\| < c$ satisfies $\varphi(t) \rightarrow x_{\text{eq}}$ as $t \rightarrow \infty$.

Definition 64. The equilibrium x_{eq} is *uniformly asymptotically stable* when it is uniformly stable (definition 61) and there exists $c > 0$ (independent of t_0) such that for each $\eta > 0$ there is $T = T(\eta) > 0$ with

$$\|\varphi(t) - x_{\text{eq}}\| < \eta \quad \forall t \geq t_0 + T(\eta), \quad \forall \|\varphi(t_0) - x_{\text{eq}}\| < c, \quad \forall t_0 \geq 0.$$

Definition 65. The equilibrium x_{eq} is *globally uniformly asymptotically stable* when it is uniformly stable with $\delta(\varepsilon) \rightarrow \infty$ as $\varepsilon \rightarrow \infty$, and for each pair of positive numbers η and c there is $T = T(\eta, c) > 0$ such that

$$\|\varphi(t) - x_{\text{eq}}\| < \eta \quad \forall t \geq t_0 + T(\eta, c), \quad \forall \|\varphi(t_0) - x_{\text{eq}}\| < c, \quad \forall t_0 \geq 0.$$

Definition 66. The equilibrium x_{eq} is *exponentially stable* when there exist positive constants c , k , and λ such that

$$\|\varphi(t) - x_{\text{eq}}\| \leq k \|\varphi(t_0) - x_{\text{eq}}\| e^{-\lambda(t-t_0)}$$

for all $t \geq t_0 \geq 0$ and all trajectories φ with $\|\varphi(t_0) - x_{\text{eq}}\| < c$.

Definition 67. The equilibrium x_{eq} is *globally exponentially stable* when the bound in definition 66 holds for every initial state $\varphi(t_0) \in \mathbb{R}^n$, i.e., $c = \infty$.

4.5 Uniform decay envelopes

A uniform convergence-time hypothesis says that for each tolerance η there is a delay $T(\eta)$ after which every trajectory is inside the η -ball. Read the other way round, it bounds how far a trajectory can still be after a given elapsed time — but only through a function of η that may be discontinuous, and so is not yet a comparison function.

The two results below do the regularization: they turn the delay into a genuine decay envelope, a function of elapsed time alone that is continuous, positive and vanishing at $+\infty$. That envelope is the time-decay half of the class $\mathcal{KL}$ bounds of the next section; the class $\mathcal{K}$ bound supplies the other half, the dependence on the initial deviation.

4.6 Comparison-function characterizations

Each stability notion above is equivalent to the existence of a comparison-function bound. These equivalences are what let the Lyapunov theorems conclude a stability property by exhibiting a bound, and conversely let a stability hypothesis be used as an estimate. Both sides quantify over the same trajectories — those defined on $[t_0, \infty)$ — so each is a restatement of one class of solutions, not a relation between two.

Theorem 68. The equilibrium x_{eq} of $\dot{x} = f(t, x)$ is uniformly stable (definition 61) if and only if there exist a class $\mathcal{K}$ function α and a positive constant c , independent of t_0 , such that

$$\|\varphi(t) - x_{\text{eq}}\| \leq \alpha(\|\varphi(t_0) - x_{\text{eq}}\|) \quad \forall t \geq t_0 \geq 0, \quad \forall \|\varphi(t_0) - x_{\text{eq}}\| < c.$$

Proof.

□

Theorem 69. *The equilibrium x_{eq} is uniformly asymptotically stable (definition 64) if and only if there exist a class $\mathcal{KL}$ function β and a positive constant c , independent of t_0 , such that*

$$\|\varphi(t) - x_{\text{eq}}\| \leq \beta(\|\varphi(t_0) - x_{\text{eq}}\|, t - t_0) \quad \forall t \geq t_0 \geq 0, \forall \|\varphi(t_0) - x_{\text{eq}}\| < c.$$

Proof.

□

Chapter 5

Class $\mathcal{KL}$ functions

5.1 Inverses of antitone functions

A function W that is continuous and strictly antitone on $(0, \infty)$, blows up at 0^+ and vanishes at $+\infty$ surjects onto $(0, \infty)$, and its right inverse has the same shape. This is the mechanism behind the class $\mathcal{KL}$ construction of the next section: W is a mean-value bound and its inverse supplies the decay rate.

Theorem 70. *Let $W : \mathbb{R} \rightarrow \mathbb{R}$ be continuous on $(0, \infty)$ with $W(\eta) \rightarrow +\infty$ as $\eta \rightarrow 0^+$ and $W(\eta) \rightarrow 0$ as $\eta \rightarrow +\infty$. Then for every $s > 0$ there exists $c > 0$ with $W(c) = s$.*

Proof. □

Theorem 71. *Under the surjectivity conditions of theorem 70, $W(\text{invFunOn } W(0, \infty) s) = s$ for all $s > 0$.*

Proof. □

Theorem 72. *Under the surjectivity conditions of theorem 70, $\text{invFunOn } W(0, \infty) s > 0$ for all $s > 0$.*

Proof. □

Theorem 73. *If W is strictly antitone on $(0, \infty)$, then $\text{invFunOn } W(0, \infty)$ is strictly antitone on $(0, \infty)$.*

Proof. □

Theorem 74. *Under the conditions of theorem 70 and with W strictly antitone, $\text{invFunOn } W(0, \infty) s \rightarrow 0$ as $s \rightarrow +\infty$.*

Proof. □

5.2 Class $\mathcal{KL}$ functions and the Osgood construction

Definition 75. A *class $\mathcal{KL}$ function* on $[0, a) \times [0, \infty)$ is continuous, class $\mathcal{K}$ in the first argument, and for each fixed $r > 0$ is strictly decreasing and tends to 0 as $s \rightarrow \infty$. It arises as the bound $\|x(t)\| \leq \beta(\|x_0\|, t)$ in asymptotic stability estimates.

Definition 76. A *global class $\mathcal{KL}$ function* is a map $\beta : [0, \infty) \times [0, \infty) \rightarrow \mathbb{R}$ that is class $\mathcal{K}_\infty$ in the first argument (continuous, strictly increasing, zero at zero, radially unbounded) and, for each fixed $r \geq 0$, is antitone and tends to 0 as $s \rightarrow \infty$.

5.2.1 Building class $\mathcal{KL}$ functions

Two constructions and the two composition rules, in the bounded and the global setting. The product $\alpha(r)\gamma(s)$ of a class $\mathcal{K}$ and a class $\mathcal{L}$ function is the easy one; the Sontag-style cap is the one that does real work.

The cap is built in the form $\min(\alpha(r), \sqrt{\alpha(r)U(s)})$, so the decay of that particular expression is recorded first.

5.2.2 Smoothing a bound into a class $\mathcal{KL}$ function

A candidate bound that has the right qualitative behaviour — monotone in the initial deviation, decaying in time — need not itself be class $\mathcal{KL}$: it may fail continuity or strict monotonicity. These two results replace such a candidate by a genuine class $\mathcal{KL}$ function dominating it, which is how the stability characterizations produce their bounds.

5.2.3 The decay ODE and the comparison bound

A class $\mathcal{K}$ decay rate produces a class $\mathcal{KL}$ bound. This is the step that turns a Lyapunov estimate — which controls only the *rate* of decrease — into an asymptotic stability statement, and it is where the scalar ODE $\dot{y} = -\alpha(y)$ enters.

Theorem 77. *Osgood construction.* *Given a class $\mathcal{K}$ function α satisfying the Osgood condition $\int_0^\varepsilon \frac{1}{\alpha(x)} dx = +\infty$, the function $\sigma(r, s) = \eta^{-1}(\eta(r) + s)$, where $\eta(y) = -\int_{\text{base}}^y \frac{1}{\alpha(x)} dx$, is a class $\mathcal{KL}$ function (definition 75).*

Proof.

□

5.3 Assumed comparison-function results

The three statements below are **axioms** in the Lean development: they are assumed, not proved, and every result that depends on one inherits that assumption. They are standard comparison-function facts, the first being the global analogue of Lemma 9 of Kellett’s compendium. Discharging them is open work.